Project – Data Cleaning and Analysis with Power Query

In this project, I used various features of **Power Query** to clean and preprocess a dataset for analysis. The objective was to ensure data consistency, remove unnecessary formatting issues, and create new calculated columns to support downstream insights. Below are the step-by-step actions I performed:

1. I selected the raw data range and used **Data > From Table/Range** to load the dataset into Power Query. I confirmed the table creation with headers by clicking **OK**.
2. To remove redundancy, I used **Remove Rows > Remove Duplicates** to eliminate any duplicate records in the dataset.

```
= Table.Distinct(#"Changed Type", {"Order ID"})
```

	Order ID	Order Date	Arrival Date	Name	Department
1	25691	3/8/2024 12:00:00 AM	3/16/2024 12:00:00 AM	Bill SmITH	Cloud Tech

3. To compute the **number of days between the order date and arrival date**, I selected both columns (first Arrival Date, then Order Date) and navigated to **Add Column > Date > Subtract Days**. This generated a new column with the calculated duration in days.

```
= Table.AddColumn(#"Renamed Columns", "Duration", each [Arrival Date]-[Order Date])
```

	Department, Region	Payment Type	Revenue	Profit	Duration
1	d Tech, Texas	Card	42845	8846	8.00:00:00

```
= Table.TransformColumns(#"Added Custom",{{"Duration", Duration.Days, Int64.Type}})
```

	Department, Region	Payment Type	Revenue	Profit	Duration
1	d Tech, Texas	Card	42845	8846	8
2	tegy, New York	Card	34152	13489	84

4. The **Name** column was cleaned by applying two formatting steps:
   a) **Transform > Format > Capitalize Each Word** ensured consistent capitalization of names.

```
= Table.TransformColumns(#"Replaced Value1",{{"Name", Text.Proper, type text}})
```

	Name	Department	Region	Payment Type	Revenue
1	Bill Smith	Cloud Tech	Texas	Card	42
2	Ken Singh	Strategy	New York	Card	34
3	Harley Fritz	Strategy	New York	Card	15

   b) **Transform > Format > Trim** was used to remove any leading or trailing spaces.

```
= Table.TransformColumns(#"Capitalized Each Word",{{"Name", Text.Trim, type text}})
```

	Name	Department	Region	Payment Type	Revenue
1	Bill Smith	Cloud Tech	Texas	Card	42
2	Ken Singh	Strategy	New York	Card	34

5. To fix extra spacing between first and last names, I selected the Name column, chose **Replace Values**, and replaced **double spaces** with a **single space**. This corrected the formatting issue and improved readability.

```
= Table.ReplaceValue(#"Trimmed Text","  "," ",Replacer.ReplaceText,{"Name"})
```

	Name	Department	Region	Payment Type	Revenue
1	Bill Smith	Cloud Tech	Texas	Card	42
2	Ken Singh	Strategy	New York	Card	34

6. For better segmentation, I merged the **Department** and **Region** columns into a single column using **Transform > Merge Columns**, with a comma followed by a space as the separator.

Prepared By: Nazrana Haque
nazrana@gmail.com

```
= Table.CombineColumns(#"Replaced Value4",{"Department", "Region"},Combiner.CombineTextByDelimiter(",
    ", QuoteStyle.None),"Merged")
```

	Order ID	Order Date	Arrival Date	Name	Merged
1	25691	3/8/2024	3/16/2024	Bill Smith	Cloud Tech, Texas
2	27870	1/7/2024	3/31/2024	Ken Singh	Strategy, New York

7. The **Payment Type** column contained inconsistent labels like *Credit Card* and *Card*, or *Transfer* and *Bank Transfer*. I replaced to categorize these into unified labels.

```
= Table.ReplaceValue(#"Renamed Columns","Credit Card","Card",Replacer.ReplaceText,{"Payment Type"})
```

	Order Date	Arrival Date	Name	Department, Region	Payment Type
1	3/8/2024	3/16/2024	Bill Smith	Cloud Tech, Texas	Card
2	1/7/2024	3/31/2024	Ken Singh	Strategy, New York	Card
3	7/9/2024	8/8/2024	Harley Fritz	Strategy, New York	Card
4	7/31/2024	9/16/2024	Nyla Novak	Operations, Florida	Card
5	12/30/2023	3/18/2024	David Rasmussen	Operations, Florida	Transfer

8. I calculated the **Profit Margin** by creating another **Custom Column** using the formula: Profit / Revenue. This newly calculated column was then formatted as a **percentage** using the percentage number format option.

```
= Table.AddColumn(#"Reordered Columns", "Profit Margin", each [Profit]/[Revenue])
```

	Department, Region	Payment Type	Revenue	Profit	Profit Margin
2	tegy, New York	Card	34152	13489	0.394969548
3	tegy, New York	Card	15829	67998	4.295786215

```
= Table.TransformColumnTypes(#"Added Custom1",{{"Profit Margin", Percentage.Type}})
```

	Department, Region	Payment Type	Revenue	Profit	Profit Margin
1	d Tech, Texas	Card	42845	8846	20.65%
2	tegy, New York	Card	34152	13489	39.50%

9. To perform aggregation analysis, I **duplicated the original query** (right-clicked on Table1 > Duplicate), renamed it for clarity, and used **Home > Group By > Advanced** to summarize the data based on selected fields.

Queries [3]
- Table1
- Payment Analysis
- Days Analysis

```
= Table.Group(#"Changed Type2", {"Payment Type"}, {{"Avg Rev", each List.Average([Revenue]), type
    nullable number}, {"Avg Profit Margin", each List.Average([Profit Margin]), type nullable number},
    {"Count", each Table.RowCount(_), Int64.Type}})
```

	Payment Type	Avg Rev	Avg Profit Margin	Count
1	Card	32010.64286	0.798848368	14
2	Transfer	33417.09412	0.511933277	17

Queries [3]
- Table1
- Payment Analysis
- Days Analysis

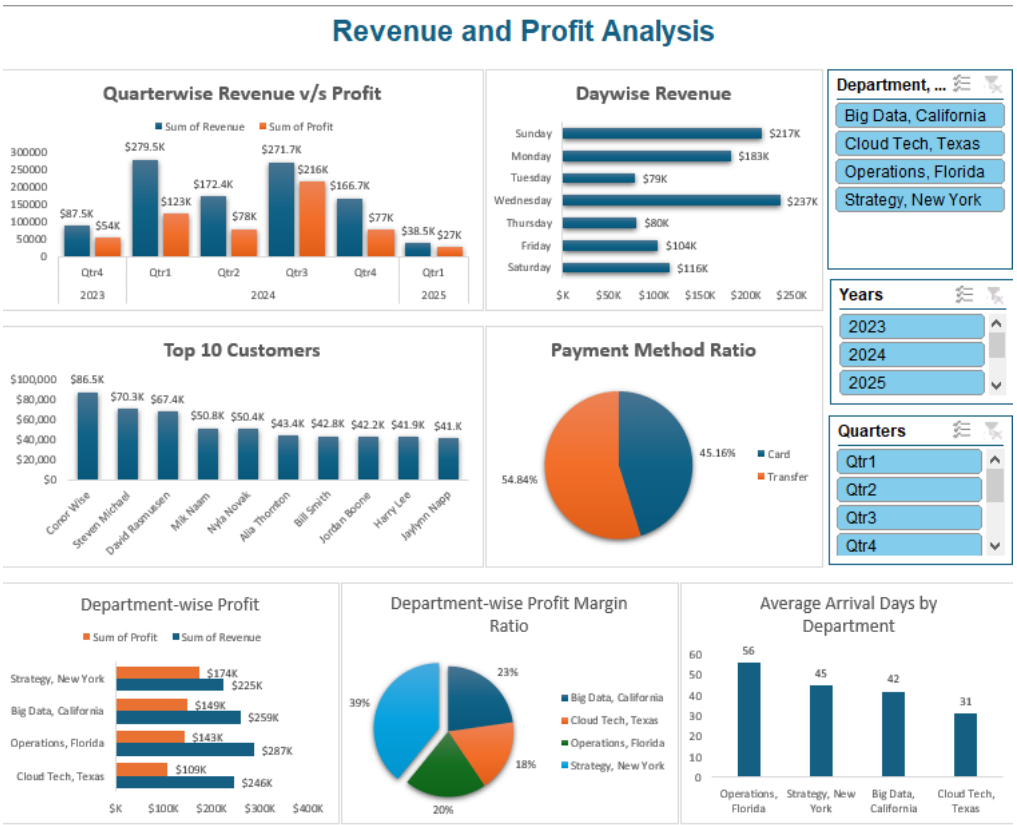
```
= Table.Group(#"Changed Type2", {"Department, Region"}, {{"Avg Days", each List.Average([Duration]),
    type number}, {"Min", each List.Min([Duration]), type number}, {"Max", each List.Max([Duration]),
    type number}})
```

	Department, Region	Avg Days	Min	Max
1	Cloud Tech, Texas	31.42857143	8	58
2	Strategy, New York	44.75	9	84
3	Operations, Florida	54.71428571	3	79
4	Big Data, California	42	8	84

10. After completing all transformations, I selected **Close & Load** to load the cleaned and prepared dataset back into Excel, making it ready for further analysis or reporting.

Prepared By: Nazrana Haque
nazrana@gmail.com

A	B	C	D	E	F	G	H	I	J
Order ID	Order Date	Arrival Date	Days to A	Name	Department, Region	Payment Type	Revenue	Profit	Profit Margin
25691	3/8/2024	3/16/2024	8	Bill Smith	Cloud Tech, Texas	Card	42845	8846	0.206465165
27870	1/7/2024	3/31/2024	84	Ken Singh	Strategy, New York	Card	34152	13489	0.394969548
46726	7/9/2024	8/8/2024	30	Harley Fritz	Strategy, New York	Card	15829	67998	4.295786215
35895	7/31/2024	9/16/2024	47	Nyla Novak	Operations, Florida	Card	50373	21335	0.423540389
27303	12/30/2023	3/18/2024	79	David Rasmussen	Operations, Florida	Transfer	46467	23668	0.509350722
25985	10/1/2024	10/25/2024	24	Ivan Hiney	Cloud Tech, Texas	Transfer	35036	9788	0.279369791
46060	8/4/2024	8/18/2024	14	Jonha Ma	Cloud Tech, Texas	Card	35205	24643.5	0.7
32107	2/3/2024	3/4/2024	30	Jordan Boone	Cloud Tech, Texas	Transfer	42246	29572.2	0.7
14750	4/8/2024	5/30/2024	52	Kylee Townsend	Cloud Tech, Texas	Transfer	36906	20225	0.548013873
27691	9/16/2024	11/13/2024	58	Nora Rollins	Cloud Tech, Texas	Transfer	14894	7988	0.536323352
47557	4/4/2024	5/8/2024	34	Brendan Wallace	Cloud Tech, Texas	Transfer	38608	7721.6	0.2
46669	1/21/2024	4/5/2024	75	Conor Wise	Operations, Florida	Transfer	46329.6	9265.92	0.2
38741	9/23/2024	12/10/2024	78	Steven Michael	Big Data, California	Transfer	21889	13107.5	0.598816757
18517	3/15/2024	5/1/2024	47	Lucia Mckay	Big Data, California	Card	18249	11589	0.635048496
15314	12/1/2024	1/19/2025	49	Jose Roach	Big Data, California	Card	39047	20119	0.515250852
16569	1/3/2024	1/30/2024	27	Franklin Wrigt	Big Data, California	Transfer	26215	13255	0.50562655
32159	7/22/2024	9/12/2024	52	Alia Thornton	Operations, Florida	Transfer	43359	22488	0.518646648
12997	7/8/2024	9/12/2024	66	Denzel Flores	Operations, Florida	Card	39366	26798	0.680739725
33650	7/3/2024	9/2/2024	61	Conor Wise	Operations, Florida	Transfer	40125	26800	0.667912773
28321	6/26/2024	7/4/2024	8	Steven Michael	Big Data, California	Transfer	48364	24659	0.509862708
35767	3/8/2024	5/31/2024	84	Bruno Cordova	Big Data, California	Card	15097	7899	0.523216533
38601	10/15/2023	10/29/2023	14	Jaylynn Napp	Big Data, California	Card	41034	29898	0.728615295
30084	2/13/2024	4/2/2024	49	Bruce Rich	Big Data, California	Transfer	27775	14697	0.529144914
24370	1/1/2025	1/23/2025	22	Arturo Moore	Big Data, California	Transfer	21230	13500	0.635892605

Days Analysis | Payment Analysis | **Cleaned Data** | Raw Data

11. Using the Cleaned Data, I created different chats for the Dashboard.